Southern New Hampshire University

CS 505 Module Three Activity Template

Complete this template by replacing the bracketed text with the relevant information.

Constructing a Proof

1. Let N be an integer. Write a proof demonstrating that N is odd if and only if N^3 is odd.
   $N\ is\ odd \rightarrow N^3 is\ odd$
   $Assume\ N\ is\ odd.$
   $By\ definition\ of\ odd\ integers, N = 2k + 1\ for\ some\ integer\ k.$
   $N^3 = (2k + 1)^3$
   $= (2k + 1)(2k + 1)(2k + 1)$
   $= (4k^2 + 4k + 1)(2k + 1)$
   $= 8k^3 + 8k^2 + 2k + 4k^2 + 4k + 1$
   $= 8k^3 + 12k^2 + 6k + 1$
   $= 2(4k^3 + 6k^2 + 3k) + 1$

$Since\ 4k^3 + 6k^2 + 3k\ is\ an\ integer, N^3 has\ the\ form\ 2m + 1,$
$where\ m\ is\ an\ integer.\ Therefore, N^3 is\ \boldsymbol{odd}$

   $\boldsymbol{To\ prove\ the\ converse, we\ prove\ the\ contrapositive}$
   $Assume\ N\ is\ even$
   $By\ definition\ of\ even\ integers\ N = 2k$
   $N^3 = (2k)^3$
   $= 8k^3$
   $= 2(4k^3)$

$Since\ 4k^3\ is\ an\ integer, N^3\ has\ the\ form\ 2m\ where\ m\ is\ an\ integer. Therefore, N^3 is\ \boldsymbol{even}$

2. Consider the following statement: "For all integers a and b, if $a - b$ is odd, then a and b are odd." Address the following items:
   A. Write the contrapositive of the statement.
      $If\ a\ and\ b\ are\ not\ both\ odd, then\ a - b\ is\ not\ odd.$
   B. Write the converse of the statement.
      $For\ all\ integers\ a\ and\ b, if\ a\ and\ b\ are\ odd, then\ a - b\ is\ odd.$
   C. Write the negation of the statement.
      $There\ exists\ integers\ a\ and\ b\ such\ that\ a - b\ is\ odd\ and\ a\ and\ b\ are\ not\ both\ odd.$
   D. Is the contrapositive of the original statement true or false? Prove your answer.
      $The\ contrapositive\ is\ false. Consider\ a = 2\ and\ b$
      $= 1. Since\ a\ is\ even, the\ hypothesis\ is\ true.$
      $However$:
      $a - b = 2 - 1 = 1$
      $Since\ 1\ is\ an\ odd\ integer, not\ even, the\ conclusion\ is\ \boldsymbol{false}.$

$Since\ the\ hypothesis\ is\ true\ and\ the\ conclusion\ is\ false, the\ contrapositive\ is\ \boldsymbol{false}$

E. Is the converse of the original statement true or false? Prove your answer.

The converse of the original statement is **false.**
Assume a and b are odd.
By the definition of an odd integer, $a = 2m + 1, b = 2n + 1$
$a - b = 2m + 1 - (2n + 1)$
$= 2m + 1 - 2n - 1$
$= 2m - 2n$
$= 2(m - n)$
Since $m - n$ is an integer, the conclusion results in the form $2k$
The definition of an even integer is $2k$.
Since $a - b$ has the form $2k$, it is even, not odd. Therefore, the conclusion of the converse is false while the hypothesis is true. Hence, the converse is **false**.

F. Is the negation of the original statement true or false? Prove your answer.

The negation of the original statement is **true.**
Assume $a = 2$ and $b = 1$
$a - b = 2 - 1 = 1$
Since 1 is odd, the first part of the negation is true. Additionally, a=2 is even, so a and b are not both odd. Therefore, the negation of the original statement is true.

3. Prove that for any set *S*, the power set of *S* (denoted as *P*(*S*)) always contains the empty set as an element.

Let S be any set.
By definition, P(S) is the set of all subsets of S.
The empty set has no elements. Since there are no elements in the empty set that are not also in S, the empty set is a subset of S.
Therefore:
$\emptyset \subseteq S$
Since P(S) contains all subsets of S, and $\emptyset$ is a subset of S, then:
$\emptyset \in P(S)$
Therefore, for any set of S, the power set of S always contains the empty set as an element.

4. For the following statements, state whether each is true or false. Write a proof or provide a counterexample. The sets *X*, *Y*, *Z* are subsets of a universal set *U*. Counter examples must also include the definition for *U*.

A. For all sets *X* and *Z*, either $X \subseteq Z$ or $Z \subseteq X$.

The statement is **false.**
Let:
$U = \{1, 2\}$
$X = \{1\}$
$Z = \{2\}$
Since $1 \in X \; but \; 1 \notin Z, X \nsubseteq Z$
Additionally, since $2 \in Z \; but \; 2 \notin X, Z \nsubseteq X$
Therefore $X \subseteq Z \; or \; Z \subseteq X$ is **false**.

B. $X \cap (Y \setminus Z) = (X \cap Y) \setminus (X \cap Z)$

The statement is **true.**
Assume: $x \in X \cap (Y \setminus Z)$
Then: $x \in X$ and $x \in Y \setminus Z$
By the definition of set difference: $x \in Y \text{ and } x \notin Z$
Since $x \in X \text{ and } x \in Y$:
$x \in X \cap Y$
Since $x \notin Z, x \notin X \cap Z$
Therefore:
$x \in (X \cap Y) \setminus (X \cap Z)$
Thus:
$X \cap (Y \setminus Z) \subseteq (X \cap Y) \setminus (X \cap Z)$

Now assume: $x \in (X \cap Y) \setminus (X \cap Z)$
Then: $x \in X \cap Y \text{ and } x \notin X \cap Z$
Since $x \in X \cap Y$:
$x \in X \text{ and } x \in Y$
Since $x \notin X \cap Z, x \notin Z.$
Therefore: $x \in Y \setminus Z$
Since $x \in X$: $x \in X \cap (Y \setminus Z)$
Thus: $(X \cap Y) \setminus (X \cap Z) \subseteq X \cap (Y \setminus Z)$
Since both subset directions are true:
$X \cap (Y \setminus Z) = (X \cap Y) \setminus (X \cap Z)$

C. $X \times (Y \cap Z) = (X \times Y) \cap (X \times Z)$

Assume: $(a, b) \in X \times (Y \cap Z)$
Then: $a \in X \text{ and } b \in Y \cap Z$
Since $b \in Y \cap Z$: $b \in Y \text{ and } b \in Z$
Since $a \in X \text{ and } b \in Y$: $(a, b) \in X \times Y$
Since $a \in X \text{ and } b \in Z$: $(a, b) \in X \times Z$
Therefore:
$(a, b) \in (X \times Y) \cap (X \times Z)$
Thus: $X \times (Y \cap Z) \subseteq (X \times Y) \cap (X \times Z)$

Now assume: $(a, b) \in (X \times Y) \cap (X \times Z)$
Then: $(a, b) \in X \times Y \text{and } (a, b) \in X \times Z$
Therefore:
$a \in X, b \in Y, \text{ and } b \in Z$
Since $b \in Y \text{ and } b \in Z$: $b \in Y \cap Z$
Since $a \in X$: $(a, b) \in X \times (Y \cap Z)$
Thus: $(X \times Y) \cap (X \times Z) \subseteq X \times (Y \cap Z)$

Since both subset directions are true: $X \times (Y \cap Z) = (X \times Y) \cap (X \times Z)$

5. Prove that for any set S, the power set of S (denoted as $P(S)$) always has a cardinality that is greater than the cardinality of S itself; in other words, $|P(S)| > |S|$.

   Assume S is a finite set with n elements.
   $|S| = n$
   The power set $P(S)$, by definition, is the set of all subsets of S. Each element of S has two choices: included in a subset or not included in a subset
   Therefore:
   $|P(S)| = 2^n$
   For every non-negative integer n: $2^n > n$
   Therefore:
   $|P(S)| > |S|$
   Since $|S| = n$ and $|P(S)| = 2^n$, the power set of S has greater cardinality than S, thus, for any finite set S, $|P(S)| > |S|$

6. Write a proof demonstrating the following statement: "If it is raining outside, then the ground is wet. It is raining outside. Therefore, the ground is wet."

   Assume it is raining outside. We are given the statement that if it is raining outside, then the ground is wet. Since it is raining outside, the condition of the statement is satisfied. Therefore, the ground is wet.